

Setting up our Scripting Environment

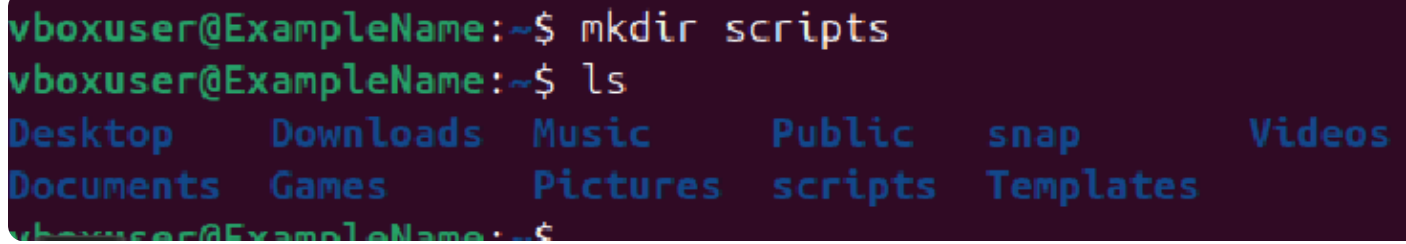
If you want to learn “bash scripting” in Linux; for automating tasks, BASH is actually the language the on you’ve typed in the *terminal* before.

Meaning, *if you’ve* done “Pre-Switch Essentials” & “Searching Lab”, those are the **BASICS of scripting**.

If not, we covered; (ls, cd, mv, cp, rmdir, mkdir, find, wc, meaning of * and ?)

As you probably guessed, scripts are FILES; and those files should be *organized somewhere*.

> **mkdir scripts** This makes the Folder that we are *putting our scripts into*.



```
vboxuser@ExampleName:~$ mkdir scripts
vboxuser@ExampleName:~$ ls
Desktop    Downloads  Music      Public     snap       Videos
Documents  Games      Pictures   scripts    Templates
```

The reason we want to ORGANIZE our scripts, is because we want to be able to run them even when we’re not inside the scripts folder.

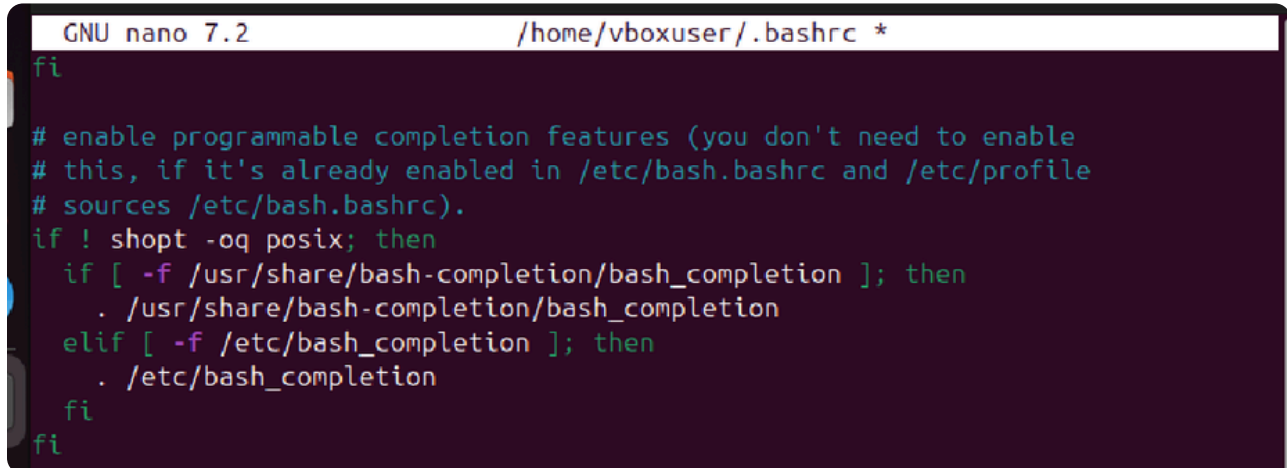
- ls, cd, mkdir are ALSO scripts we can *run from everywhere*, think of how weird it would be if we could ONLY use them from the home folder?

The PATH is something used on ALL devices to **signify** that an EXECUTABLE (like a program or a script) can be run from anywhere.

- We can ADD to this, in fact we can add *the entire folder*; so all of our scripts can be ran from anywhere.

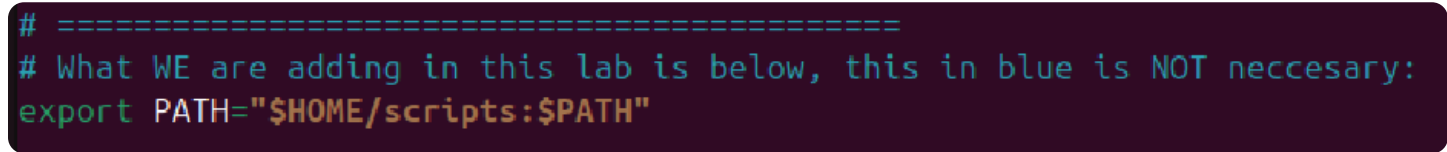
> **nano ~/.bashrc** THIS enters the **text editor** for the file *where* we can make **changes** to our terminal, the PATH being part of that.

Scroll down to the bottom, should look like this.



We are going to add this line: `export PATH="$HOME/scripts:$PATH"`

export means *add to* **PATH**, and the **"\$HOME/scripts:\$PATH"** is just our **scripts folder**.



CTRL+S to save, CTRL+X to exit.

AFTER CLOSING AND REOPENING the terminal; Now, any script we make in our folder can be used ANYWHERE.

Let's MAKE an example script to show you how to MAKE a script.

> **cd scripts/** Goes into our *scripts folder*.

> **nano example_script** Makes and opens the *text editor* for our `example_script`.

- If you have *VSCODE* installed, do **code example_script** to open it there.

FIRST, we need to tell our **script** what *language* to use.

> **#!/usr/bin/env bash** THIS is called a "shebang", it TELLS the file we are talking in the **TERMINAL language**.

Now we can just write as if we we're typing in the terminal normally, like `cd`, `ls`.

```
GNU nano 7.2                                example_script *
#!/usr/bin/env bash
date
```

date JUST shows the *date & time*, typing this in the normal terminal is the same thing. Normally you'd have MANY lines, but here it's just one.

CTRL+S to save and CTRL+X to exit

We have to make this file RUNNABLE, currently it's just a text file.

> **chmod u+x example_script** This gives the **u**ser the *ability* to **e**xecute the file.

```
vboxuser@ExampleName:~/scripts$ chmod u+x example_script
```

NOW, from anywhere we can *execute* our new script.

```
vboxuser@ExampleName:~$ example_script
Thu Jan  1 08:28:45 PM UTC 2026
```

ONE TIP going forward. When *scripting*, **first** run it through manually. Meaning, do it in the **terminal** beforehand to get the idea and *execution* down, as we will be doing this going forward.